Isha Chaudhary

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Education

University of Illinois Urbana-Champaign
Ph.D. (Dec 2026; thesis proposed) + M.S. (2025) in CS; Advisor: Gagandeep Singh; GPA: 3.97/4.00
Indian Institute of Technology Delhi, India
B.Tech in Electrical Engineering (minor in CS); GPA: 9.3/10.0; Class Rank: 1/51

Research Interests

Science of evaluation and post-training of frontier LLMs and agents: statistical model checking of agent behaviors over input distributions, automated red-teaming and failure-mode discovery, adversarial robustness of LLM judges and monitors, RL fine-tuning and adversarial post-training for agentic system alignment.

Experience

Anthropic, AI Safety Fellow *(with Robert Kirk, UK AISI; Otto Stegmaier, Anthropic)* Aug 2026 – present
- Stress-testing alignment evaluation of frontier model systems

Amazon AGI, Applied Scientist Intern *(with Rahul Gupta, Weitong Ruan)* May – Aug 2026
- Constitutional AI alignment by post-training (GRPO with Slime) with Petri over policies for frontier agent safety

Amazon Web Services, Applied Scientist Intern *(with Ganyu Teng, Shawn Zhang)* Sep – Dec 2025
- Developed a principled framework to stress-test scalable-oversight-enabled agentic systems via adversarial prompt injections, using CVaR to expose worst-case alignment failures hidden by average-case metrics
- Finetuned Qwen-1.5B with GRPO to generate prompt-injection attacks against scalable-oversight judge models

Microsoft Research, Intern *(with Ryan Beckett, Siva Kakarla, Behnaz Arzani)* May – Aug 2025
- Developed LLM-powered semantic analysis of system logs for anomaly detection, leveraging frontier model capabilities for safety-critical infrastructure monitoring

Microsoft AI (Bing), Data Science Intern May – Aug 2024
- Developed machine learning models for revenue optimization in Bing search

Adobe (Cloud Tech), Research Intern *(with Gaurav Sinha)*; US patent 2023/0274310 A1 May – Aug 2021
- Multi-task Learning with Label Proportions: Jointly disaggregated multiple aggregate labels with individual features

Selected Publications

- **Isha Chaudhary***, Vedaant Jain*, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Revealing Interpretable Failure Modes of VLMs.* Fagen workshop @ ICML 2026. (* indicates equal contribution)
- **Isha Chaudhary**, Vedaant Jain, Gagandeep Singh. *Certifying Reading Comprehension in Large Language Models.* **AISTATS 2026 Spotlight**, also in SeT LLM Workshop @ ICLR 2024.
- Chengxiao Wang, **Isha Chaudhary**, Qian Hu, Weitong Ruan, Rahul Gupta, Gagandeep Singh. *How Catastrophic is Your LLM? Certifying Risks in Conversation.* **ICLR 2026**.
  Impact: Deployed in Amazon's Nova evaluation suite
- **Isha Chaudhary**, Qian Hu, Manoj Kumar, Morteza Ziyadi, Rahul Gupta, Gagandeep Singh. *Quantitative Certification of Bias in Large Language Models.* **ICLR 2025**.
- **Isha Chaudhary**, Vedaant Jain, Avaljot Singh, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Lumos: Let there be Language Model System Certification.* PAgE workshop @ PLDI 2026.
  Impact: In deployment within National Research Platform's model cards
- Jehyeok Yeon, **Isha Chaudhary**, Gagandeep Singh. *Quantifying Distributional Robustness of Agentic Tool-Selection.* AIWILD @ ICLR 2026.
- Jason Vega*, **Isha Chaudhary***, Changming Xu*, Gagandeep Singh. *Bypassing the Safety Training of Open-Source LLMs with Priming Attacks.* **ICLR 2024** (Tiny Papers track). (* indicates equal contribution)
  Impact: Attack Success Rate ↑ 3.3× over baselines on harmful behavior benchmarks
- **Isha Chaudhary**, Alex Renda, Charith Mendis, Gagandeep Singh. *COMET: X86 Cost Model Explanation Framework.* **MLSys 2024**, also in XAI-in-Action Workshop @ NeurIPS 2023.

Awards

- Selected for **Y Combinator**'s AI Startup School (2026)
- Awarded **Best Presentation Award** (Security and Privacy track) at CSL student conference 2025 (2025)
- Awarded **Institute Silver Medal** & **Prof. C.S. Jha Memorial Excellence Award** at IIT Delhi (2022)

Talks

- **Formal Guarantees for Frontier AI Systems (NeurIPS 2026 Tutorial)**: Lead presenter
- **Certifying LLMs**: NSF NAIRR, Microsoft Research Redmond/India, IIT Delhi, Amazon-Illinois AI Center (AICE).
- **Teaching Smarter with AI**: Graduate TA training academy, CITL, UIUC.
- **Exploring COMET: Neural Cost Model Explanation Framework**: Google Compiler group, VMWare Research.

Teaching and Service

- Guest lectures on formally certifying frontier models, CS 477: Formal Software Development, UIUC, Apr 2026.
- Teaching Assistant, CS 521: Trustworthy AI Systems (Advanced Topics), UIUC, Fall 2024.
- **Area Chair**: MLSys 2025; **Reviewer**: NeurIPS 2026, ICML 2026, ICLR 2025–26, COLM 2026, KDD 2026, FAccT 2025–26, AIES 2024–25, MLSys 2025, Computing Surveys 2026; **Artifact Evaluator**: VMCAI 2024.
- Organizer, Formal Methods and Software Engineering seminar, UIUC, 2024.
- Mentored undergraduate and high-school researchers on LLM evaluation projects (URAP, URSA, SIGPLAN-M, K-12 BPC at UIUC), 2023–25.